

Center correlation breaks iid intervals

01/05

The point estimate targets β_1 , but the uncertainty model changes.

Synthetic multi-center simulation; treatment effect $\beta_1=0.25$.

$$Y_{ij} = \beta_0 + \beta_1 T_{ij} + u_j + \varepsilon_{ij}$$

$$u_j \sim N(0, \tau^2), \quad \varepsilon_{ij} \sim N(0, \sigma^2)$$

$$\rho = \frac{\tau^2}{\tau^2 + \sigma^2}$$

ρ is the intraclass correlation;
center j is the inference unit.

β_1 is the treatment effect.

u_j : center-level shift shared within
center j .

ε_{ij} : individual residual noise.

Point estimates can remain centered while iid intervals lose coverage.

Grid: G=8/20/50 centers, ICC $\rho=0/.1/.3/.5$, balanced and imbalanced cluster sizes.

The sandwich variance aggregates by center

The middle term is a center-level score covariance, not a row-level residual pool.

Both intervals estimate β_1 ; only the variance object changes.

$$\hat{V}_{\text{CR}} = (X^\top X)^{-1} \left(\underbrace{\sum_g X_g^\top \hat{u}_g \hat{u}_g^\top X_g}_{\downarrow} \right) (X^\top X)^{-1}$$

Cluster meat: score contributions are summed by center before forming covariance.

$$\hat{V}_{\text{iid}} = \hat{\sigma}^2 (X^\top X)^{-1}$$

The simulation uses cluster-robust z intervals; CR2 and wild bootstrap remain planned comparisons.

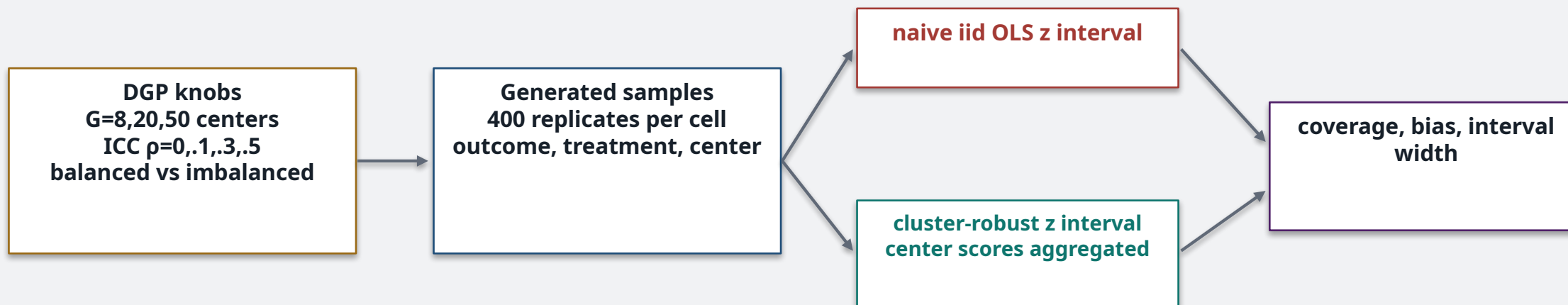
Naive variance treats rows as independent.

A stress grid separates interval behavior

03/05

The same generated samples feed two intervals and the same coverage endpoint.

Synthetic experiment: vary center count, ICC, and cluster-size balance.

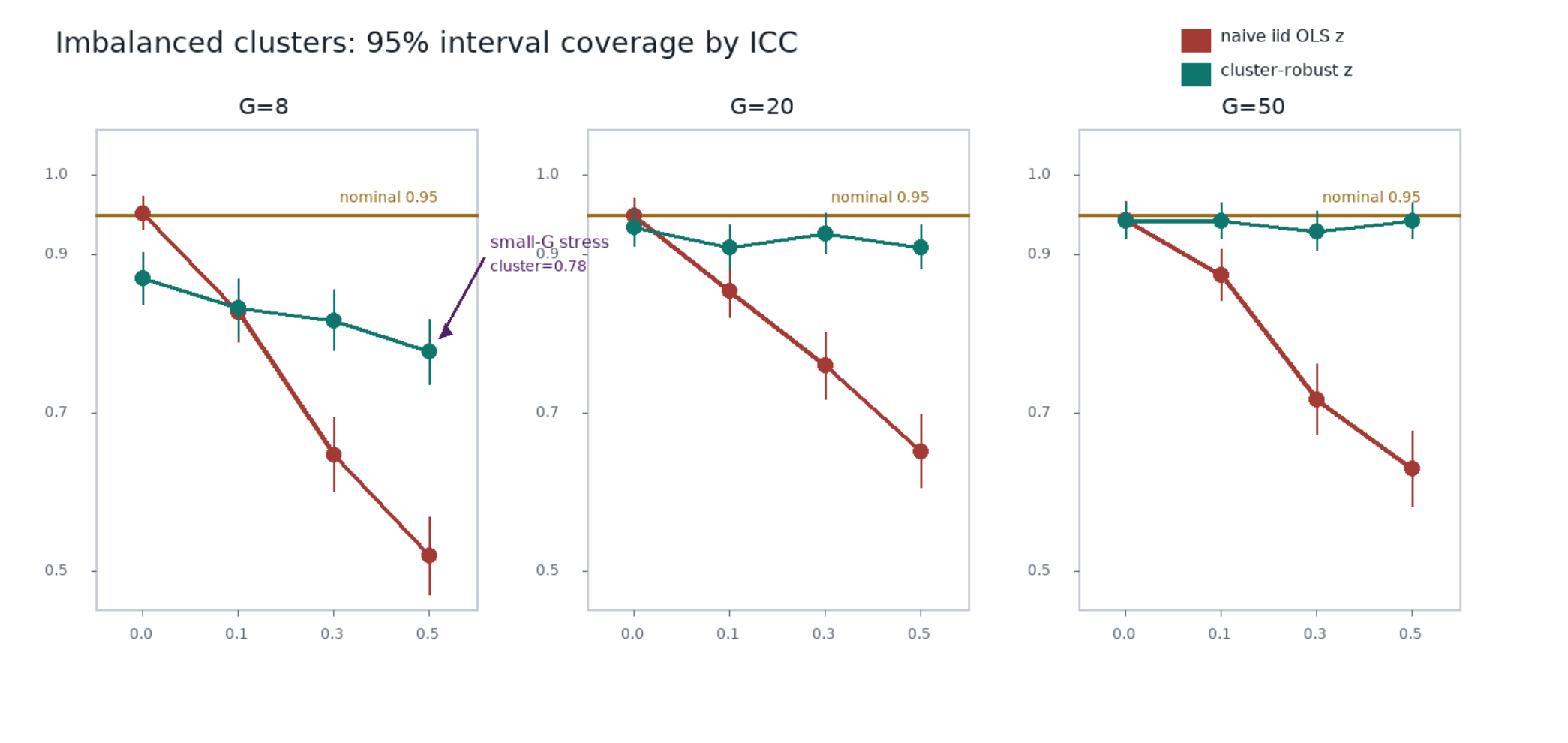


The design asks whether center aggregation repairs interval coverage across the stress grid.

Simulation study; endpoints are computed from the deterministic output for the same DGP grid.

Cluster adjustment reduces, but does not erase, undercoverage

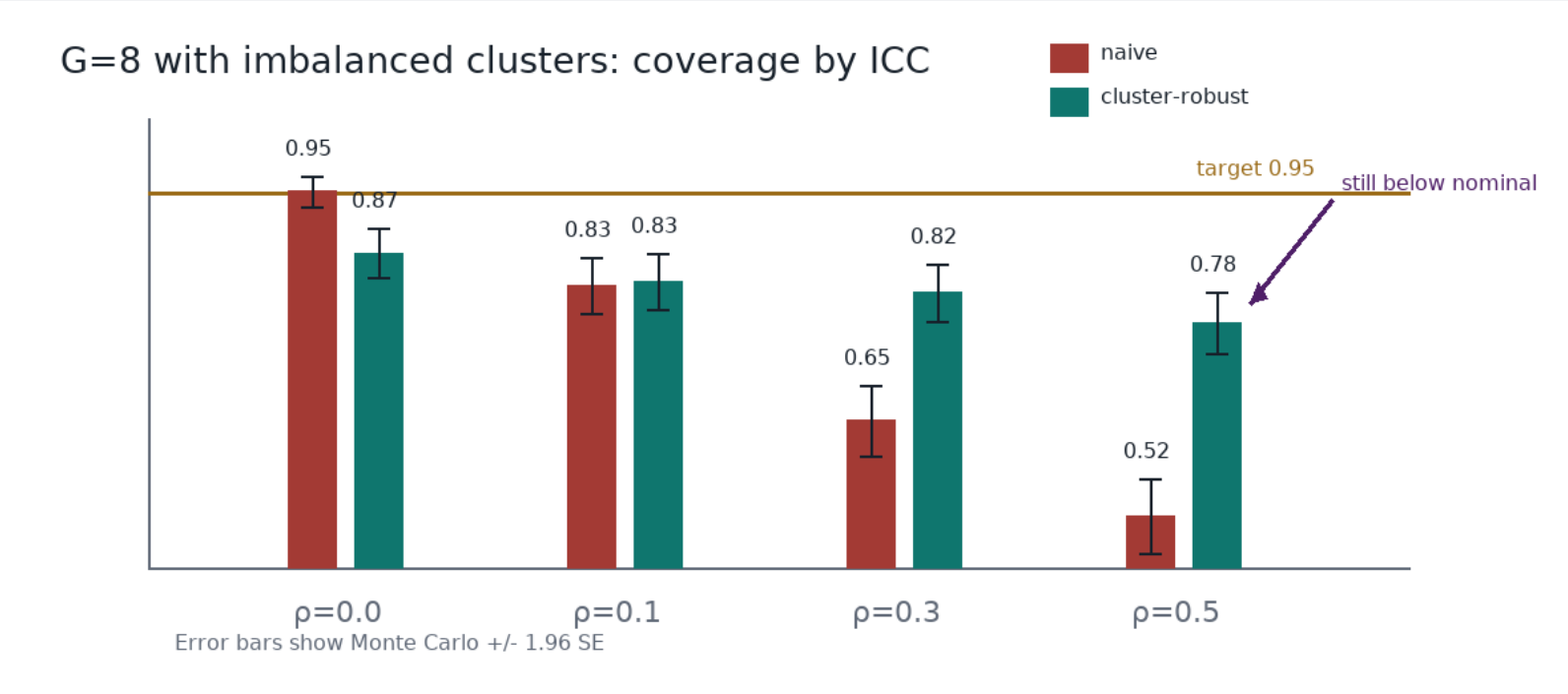
Coverage near 0.95 is the target; Monte Carlo error bars are part of the evidence.



Synthetic imbalanced-cluster simulation; vertical marks show Monte Carlo error bars.

The remaining failure is finite-center undercoverage

The next experiment tests small-sample corrections under the same grid.



At G=8, ICC $p=.5$, imbalanced clusters, cluster-robust coverage is 0.78.

Few centers make the sandwich variance noisy; imbalance increases center leverage.

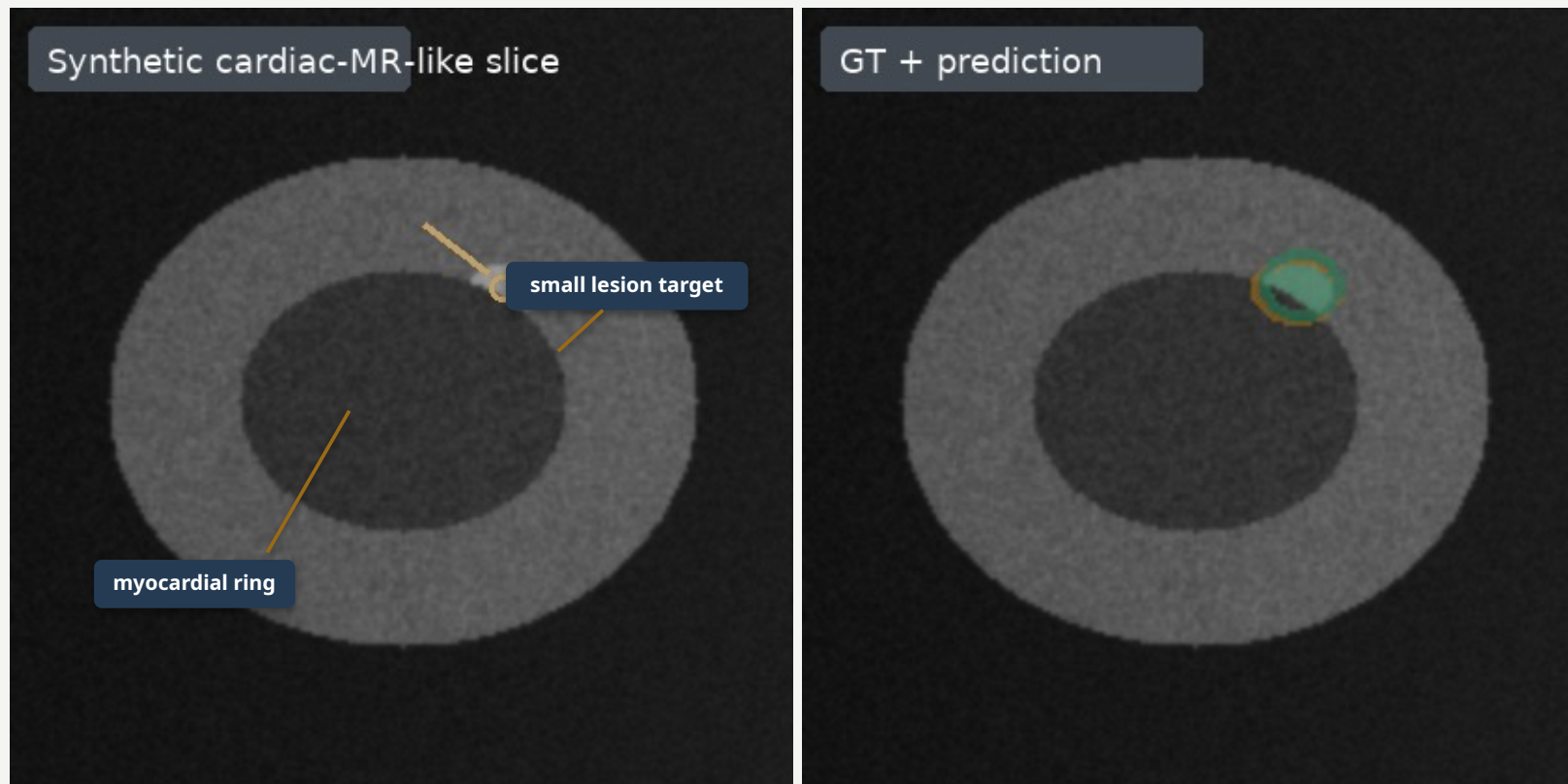
Planned comparison: CR2 small-sample correction and wild cluster bootstrap on the same grid.

Completed evidence is limited to the two intervals shown; bar intervals show Monte Carlo error bars.

A small lesion is the unit of failure

01/05

Synthetic cardiac-MR-like segmentation task; not clinical validation



Evaluation unit: lesion detection + mask overlap

Dice summarizes mask overlap, but a missed small lesion is counted at the lesion level.

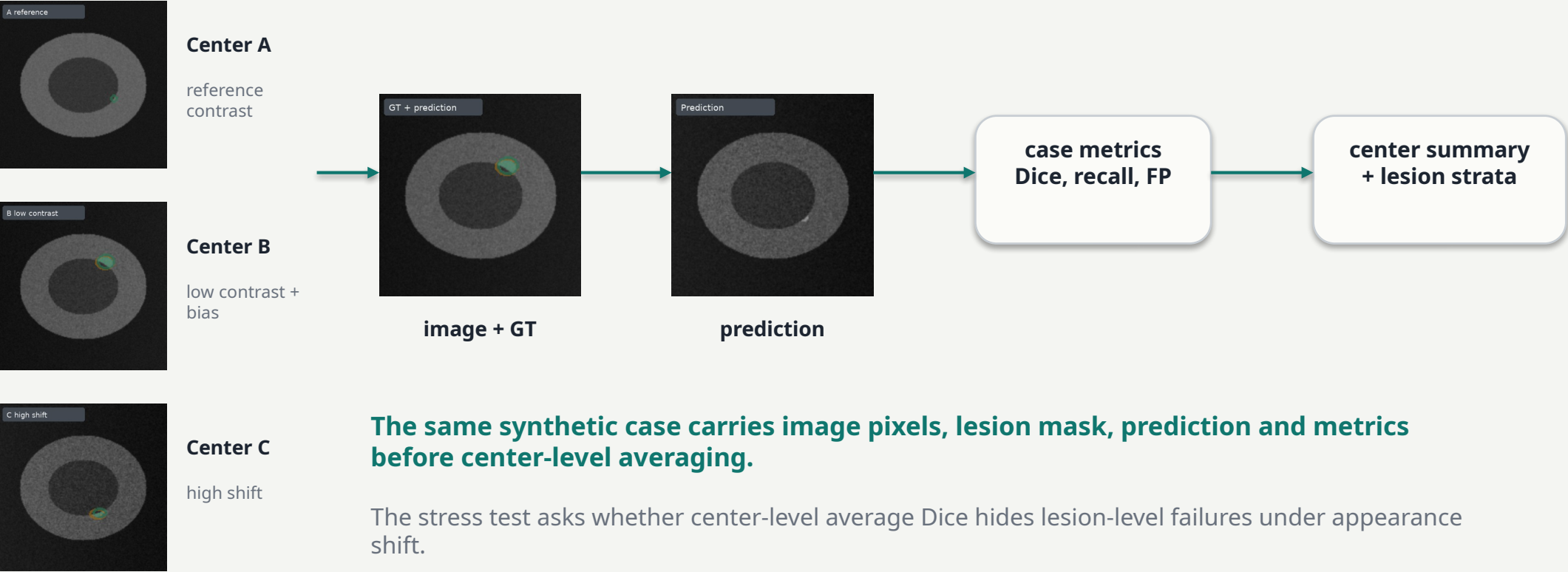
Endpoints: Dice overlap, lesion recall, false-positive burden

Synthetic cardiac-MR-like phantoms; not clinical validation.



The stress test follows one image-to-endpoint path

Center appearance shift propagates through prediction before aggregation



The same synthetic case carries image pixels, lesion mask, prediction and metrics before center-level averaging.

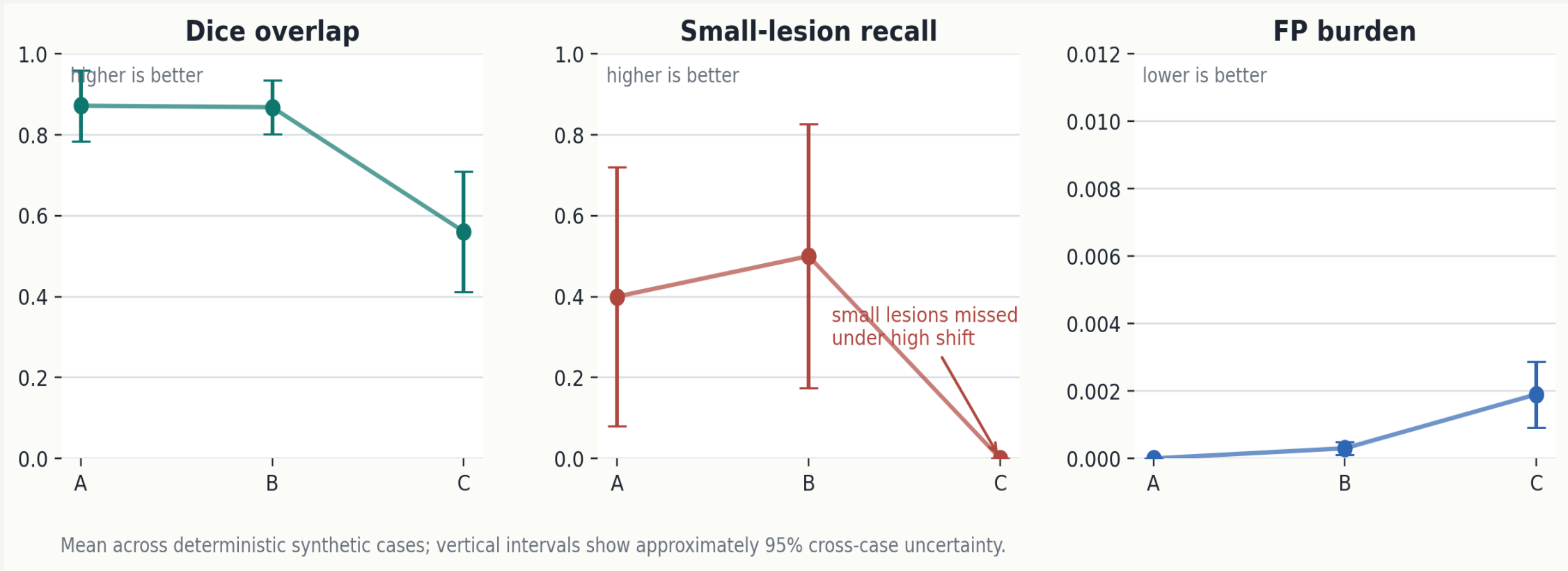
The stress test asks whether center-level average Dice hides lesion-level failures under appearance shift.

appearance shift

Aggregation is descriptive: no clinical validation claim is made from these phantoms.

Average Dice can hide a lesion-level failure

Endpoint disagreement appears under the high-shift center



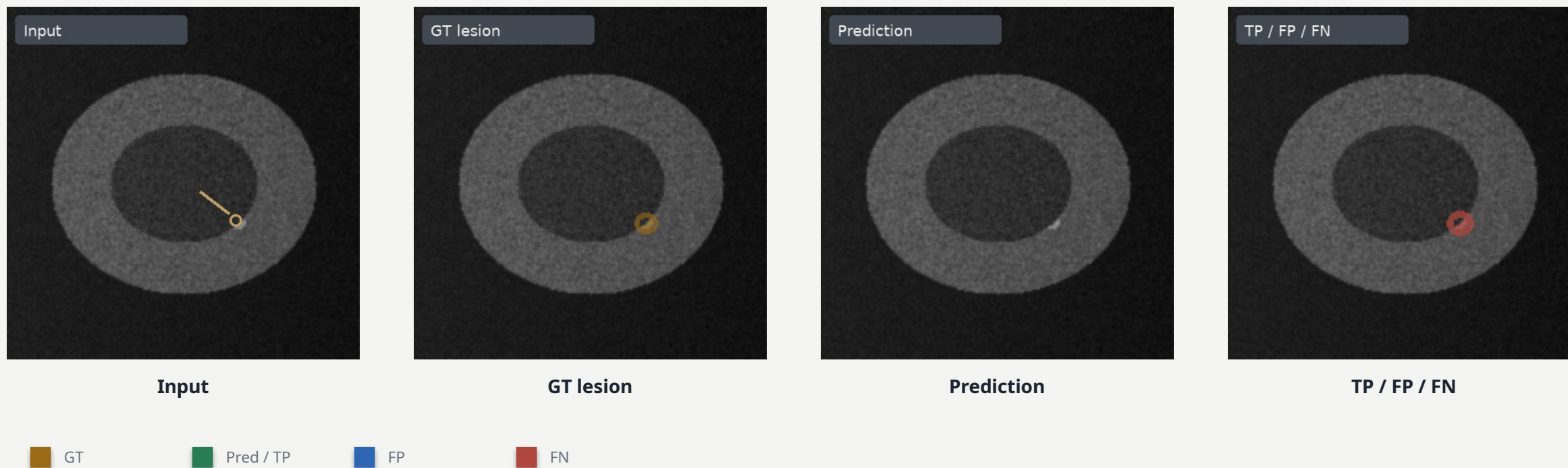
Center C average Dice = 0.56, but small-lesion recall drops to 0.00.

FP burden is lower-is-better; Dice and recall are higher-is-better.

The failure is visible in the same-case overlay

04/05

A missed small lesion drives the negative endpoint



Case C3-03 (Center C, small lesion): Dice 0.00; lesion recall 0; FN pixels 20.

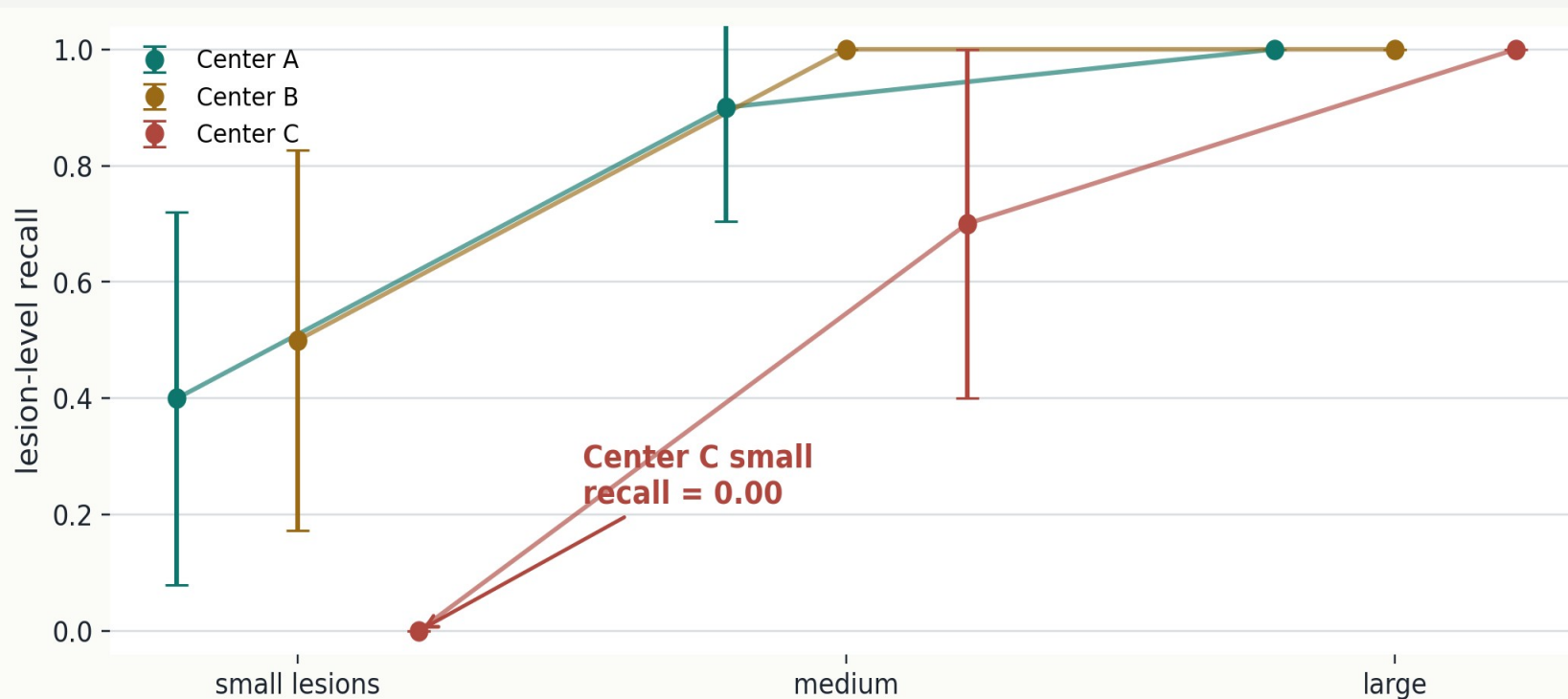
All four panels are the same synthetic slice geometry.

The red FN region shows why an average overlap metric is not enough for this stress condition.

Validate the small-lesion shift explicitly next

05/05

Completed synthetic evidence motivates a planned held-out-center experiment



Completed evidence: deterministic synthetic phantoms only; next validation is planned, not completed.

Completed evidence

Small lesions in the high-shift center are the supported failure regime in this deterministic phantom benchmark.

Planned validation

Hold out center C, stratify by lesion size, and sweep prediction threshold/calibration before comparing methods.

Synthetic cardiac-MR-like phantoms; not a patient validation endpoint.